

backslash or a quoted string does not escape this. The general form of parameter expansion is `${parameter}`, and its value is substituted.

Brackets are required in this expansion when it is a positional parameter having more than one digit or whenever it is followed by a character which is not interpreted as a part of its name. In bash, parameter expansion comes in multiple forms, and the simplest one is the dollar sign. This form of parameter expansion hardly substitutes the overall value of the variable in place of the expression.

The first character in parameter expansion is exclamation (!), and it is not named ref introducing the level of indirection. Its useful form is used by default for the variable if not set earlier, and it is done with the help of syntax `:${VAR:-DFLT}`. This can be used to allow the code to get easily modified with the help of variables. Consider the following script:

Command expansion – Command expansion permits the use of the command output in the form of an expansion. Linux possesses a utility in the command line known as `expand` through which one can easily convert tabs into spaces in the file, and when no file is defined, then it reads from the standard

```
TEST_MODE=${TEST_MODE:-0}
...
if [[ $TEST_MODE -eq 0 ]]; then
    echo "Running in live mode"
else
    echo "Running in test mode"
fi
```

output. This expansion occurs when the command is closed and represented as- `$(command)` or `'command'`. Command expansion supports the output of a command and removes the new lines during word splitting.

The expansion is performed by bash after executing the `'command'` interpretation, and it also replaces the command substitution with the standard output of command, thereby deleting any trailing lines. Command expansion can be nested and when substitution appears within double quotes then file-name expansion is not performed on the outcomes. When a standard output replaces the command that output is presumed as text and it can be assigned to any variable just like the other value. Example-